Concentrated energy does not fold space: seven pre-registered tests in a graph toy model of emergent geometry

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(Dated: DRAFT, 11 October 2026)

The author's hypothesis holds that a black hole is a region of space re-curled toward the arrangement it came from, paid for by concentrated energy. Combinatorial quantum gravity (CQG) reads it as a bubble of the random phase. We ask a graph toy: CQG's energy with the coefficient λ of its local term raised above 1 (only $\lambda = 1$ is CQG). Flat space is given energy, and vertices are classified by a count of squares as folded or melted. Seven pre-registered tests cover two, three and four directions; energy spread, packed or drained; points named or interchangeable; and, in the last, a tie between directions that the author chose. Every overall verdict is a form of MELTS or HEALS. One sub-reading is an exception by the letter: at 36 vertices, where no complete fold exists, one protocol reads as folding on three damaged replicas. Fold predictions recorded for the author failed in all seven; ours failed in part in two and missed details in two more. In three directions at moderate energies the melted states, read from the wiring, are a few flickering scars that heal when cooled. Under the author's tie, with twice the price of curling a whole direction packed into one place, half the space ends broken and none of it curled. Exactly, no single move out of a flat sheet adds a square.

I. WHAT IS ASKED

This is paper 12 of a series of short papers [1]. Companion papers study a curled torus that converts to a flat one [2], what it leaves behind [3], and how curled directions open with six and eight links [4]. This paper asks the reverse. Can energy put into flat space curl it back up?

The author's claim. A black hole is a closed region of space re-curled toward the curled arrangement that space came from. The energy concentrated in it pays the curling cost. In open space energy spreads and space heals; in a closed region space folds. Space itself may be stable: the return is paid for, not fallen into [1].

The published reading. In combinatorial quantum gravity (CQG) a black hole is a large bubble that has reverted to the random phase [5]. That work reads the random phase as matter. The author does not.

The two readings differ in what concentrated energy does to flat space: curl it, or disorder it. A sealed simulation can count both.

A benchmark. In the author's wider picture the fold is the return leg of a cycle, and new space is born where black holes collide. A published cousin is cosmological natural selection, in which black-hole singularities bounce into new expanding universes [6]. We cite it as a benchmark for what such a cycle is asked to do. Nothing here tests it or draws support from it.

Conventions. Each statement is one of four kinds, and we label it: *exact* (arithmetic or complete enumeration, at the sizes stated); *measured* (a run registered before it ran, with its verdict label in capitals); *interpretation* (ours, unverified); or *the author's* (the hypothesis's claim). Runs that were not registered are marked exploratory. Only $\lambda = 1$ is CQG. Every run of the tests here is at $\lambda > 1$, and two tests add a further term, a tie between directions. We call this family the toy. It sits next to CQG and is not CQG.

II. THE TOY, THE SEALED BOX AND THE READING RULE

State and energy. The state is a $2D$-regular two-sided (bipartite) graph on N vertices under the hard-core rule of CQG [5]. We use $D = 2$, 3 and 4: four, six and eight links per vertex, which we call two, three and four directions. With S squares (4-cycles) and $X = \sum_e (S_e - 2D + 2)_+$ surplus squares on over-full links,

$$H = 16\left[\tfrac{1}{2}D(D-1)N - S\right] + 4\lambda X. \qquad (1)$$

The flat D-torus has $H = 0$; we call it flat space. A direction of length four is called curled. *Exact:* a curled direction costs $a = 4(\lambda - 1)$ per vertex (ϵ in Ref. [2]), and the costs of several curled directions add [1, 4]. At four links Eq. (1) is the published energy with λ raised. The six- and eight-link forms are derived by us from the published definitions.

Reproduction gate, stated once. The six-link kernel has not reproduced the published six-link curve (Fig. 3 of Ref. [7]). A factor of two in the coupling is accounted for by the published equation; the remaining difference is unsettled. No published eight-link curve exists to compare with. At four links the kernel reproduces the four-link chain draw for draw, and its ladder of curled states is exact. Every six- and eight-link statement below is about our kernel and carries this open gate [1].

The reading rule. The square-count signature $d(v)$ is the number of pairs of links at v that close no square. It

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is D in flat space, falls by one for each curled direction, and rises where squares are missing. A vertex is *folded* if $d < D$ and *melted* if $d > D$; both count as damaged. A replica, read from its final block, is FOLDED if at least four vertices are damaged and folded ones are at least as many as melted, MELTED if at least four are damaged and melted outnumber folded, and HEALED if fewer than four are damaged. Four is the smallest curled object, one column. A cell's outcome is the majority over its replicas. A test reads RE-CURLS if some cell has a FOLDED majority, MELTS if none has and some cell has a MELTED majority, and HEALS if every cell is HEALED. Two tests used other registered rules: T21 the mean folded share of the damage, and T47 shares of vertices (Sec. V). These labels are counts. The reference values alone do not establish geometry, dimensionality or a return to the random phase. Sections III and IV say what the MELTED states can be, and what they were where their wiring was read.

The sealed box. Stores of nonnegative energy sit beside the graph [8]. A proposed switch of links draws on one store, is refused if the store cannot pay, and pays into it when it lowers the energy. H plus the stores, plus whatever has been drained, is conserved by construction. Conservation is bookkeeping; it does not establish equilibrium. Three protocols differ in where the energy starts and which store a move may use. *Spread*: a shared bath of $2N$ stores with the whole budget in one, which any move may draw on. *Packed*: one store per vertex, the budget in the store of one vertex, and a move pays only from the store of the first vertex it involves; or the budget wired into the graph as disorder near one vertex, with the bath empty. *Drained*: the spread protocol with every store scaled by $1 - \ell$ after each block of 100 sweeps. The cheapest move out of flat space costs 32, 64 or 128 with four, six or eight links (Sec. VI), so a single store holding less can never act.

Named or interchangeable points. By default each vertex has its own name. In the interchangeable (unlabeled) ensemble a graph gains weight in proportion to its number A of side-preserving automorphisms, and the ratio $A(G')/A(G)$ enters the acceptance of a move [1, 9]. This can favor symmetric arrangements. It does not guarantee that a folded endpoint dominates every partial state, and it does not set a formation rate. It is a modeling choice; quantum indistinguishability does not derive it here.

Table I lists the seven tests.

III. TWO DIRECTIONS

All runs in this section are on flat four-link tori at $\lambda = 1.25$ unless stated.

Exploratory predecessor. A flat sheet of 144 or 160 vertices was given 0.5 to 16 units per vertex in a single store, at $\lambda = 1.05$ to 2.0. Melted vertices outnumbered folded ones at every budget and every λ. The prediction written before the run, that small budgets buy folds because a fold is the cheaper damage, failed. The author's correction to the first reading of this run stands: heating a sheet at equilibrium randomizes it by construction. What the run tests is narrower: two ways of holding the same energy, of which folding lost [1].

Spread (T21), measured. $N = 36$ and 64; budgets of 2, 4, 8 and 16 per vertex through the bath and 0.5 to 4 through a single store; six replicas per cell; named and interchangeable points; 4,000 sweeps. At $N = 64$, where complete folds fit (four 4-cubes, or the 16×4 curled torus), the mean folded share of the damage under the bath is at most 0.27 with interchangeable points and 0.18 with named ones, at budget 2, and falls to 0.02 at 16. The registered overall reading is the one at $N = 64$ under the bath: MELTS EITHER WAY. The single store reads MELTS EITHER WAY at both sizes. At $N = 36$, where no complete fold exists, the bath reads FOLDS WITH INTERCHANGEABLE POINTS by the letter: 0.61 against a named 0.43 at budget 2, from three damaged replicas, with the order reversed at budget 4 (0.25 against 0.40). The author's statement that a closed region folds corresponds to FOLDS WITH INTERCHANGEABLE POINTS; she had given no separate pick. It failed at $N = 64$. Ours, MELTS EITHER WAY at both sizes, held at 64 and failed at 36 under the bath.

The author's objection to the spread protocol: a black hole is energy packed into one place, not energy spread evenly.

Packed (T26), measured. Twenty-seven cells: 32 to 192 units in one vertex's store at $N = 144$ and 256; 288 and 576 units over nine stores at $N = 144$; a disordered patch of 32 to 128 units in a cold shared bath at $N = 144$ and 256, and with per-vertex stores at $N = 144$; and at $N = 64$ a patch of 32 or 64 units with named and with interchangeable points. Twenty replicas per cell and 50,000 sweeps (ten and 4,000 at $N = 64$). With the budget in one store, 19 or 20 of 20 replicas are MELTED in every cell; over nine stores, 20 of 20. A patch in a shared cold bath: 15 to 20 of 20 MELTED, the rest HEALED, at most one FOLDED. A patch with per-vertex stores: 20 of 20 MELTED. At $N = 64$ the named patch is MELTED in 7 of 10 at both budgets, with 3 FOLDED at 32; the interchangeable one is HEALED in 10 of 10 at 32 and MELTED in 7 of 10 at 64. No cell has a FOLDED majority. The largest folded piece in any final state is 11 vertices, and at most 1 replica in 20 melts first and folds later. Verdict: MELTS, with 26 cells MELTED, one HEALED and none FOLDED. The author's prediction, RE-CURLS with melt-then-fold common, was inferred by the assistant and confirmed by her before any cell was read as a verdict. It failed. Ours, MELTS, held, with two misses: we expected the patch in a shared cold bath to heal at 144 and 256 vertices, and it mostly did not; and we expected the two kinds of point to agree at $N = 64$, and they differed at 32 units.

What MELTED can mean at the smallest budgets.

TABLE I. The seven pre-registered tests. D is the number of directions ($2D$ links per vertex). Verdict labels are the registered ones. "Inferred" means inferred for the author by the assistant from her stated positions.

Test	D	N	λ	Protocol; points	Verdict	Predicted for the author	Ours
T21	2	36, 64	1.25	spread; named and interchangeable	MELTS EITHER WAY (overall, $N = 64$); at $N = 36$, bath, FOLDS WITH INTERCHANGEABLE POINTS by the letter	a closed region folds (inferred from her statement): failed at 64	MELTS EITHER WAY: held at 64, failed at 36 (bath)
T26	2	64, 144, 256	1.25	packed; named, and both at 64	MELTS	RE-CURLS (inferred, then confirmed by her): failed	MELTS: held
T27	2	144	1.25	spread, then drained; named	STAYS MELTED	FOLDS BEFORE IT FLATTENS (inferred): failed	STAYS MELTED: held
T34	3	216, 512	1.25	packed and spread; named	MELTS	RE-CURLS (her words): failed	MELTS: held
T42	3	512	1.02	spread; interchangeable, with a named control	HEALS; control MELTS	RE-CURLS, control MELTS (inferred): failed; held	MELTS in both: failed for interchangeable
T47 B	4	2,304	1.30	packed; named; a tie table (Sec. V)	HEALS; control HEALS	FRONT, control HEALS (inferred, then confirmed by her): failed; held	HEALS in both: held
T57	3	216, 512	1.10, 1.25	packed and spread; named; the author's tie (Sec. VII), and untied	MELTS, tied and untied	RE-CURLS with the tie (her words): failed	MELTS or HEALS, tied and untied: held

Exact: the cheapest move out of a flat sheet loses two squares and costs 32. It leaves eight vertices at $d = 3$, as the registration states and its disclosed smoke run showed. By the rule, a replica that buys that one move and nothing more has eight damaged vertices, none folded, and reads MELTED. So at the smallest packed budgets the label can mean one broken-square defect, not a disordered region. The two-direction end states were classified by the count; our record holds no reading of their wiring of the kind given in Sec. IV.

Drained (T27), measured. $N = 144$; the spread protocol at 2, 4 and 8 per vertex; drained at $\ell = 0.001$, 0.01, 0.1 and 1.0 per block, the last a quench; a sealed control; twelve replicas per cell; 30,000 sweeps. All fifteen cells, the control included, have a MELTED majority. Of the 144 draining replicas 134 are MELTED and 10 FOLDED. No replica melts first and folds later, and the largest folded piece anywhere is 16 vertices. The quench freezes the disorder as it stands, and the slow drains do not anneal it within the run. Verdict: STAYS MELTED. The prediction inferred for the author by the assistant, FOLDS BEFORE IT FLATTENS, failed; the record does not show her confirming it. Ours, STAYS MELTED, held as a verdict; we had expected the slow drains to flatten the sheet, and they did not.

Left alone (exploratory). With a chain that also allows triangles and pentagons, run at fixed coupling g and not sealed, a flat sheet at $g = 1$, 2 and 4 accepted no move at all, and at $g = 8$ it took up 7.9 units per vertex as disorder. It did not fold. A third part of that run, an apparent jammed state, was withdrawn as an artifact of a slow chain [1].

IV. THREE DIRECTIONS

The author's mechanism. The collapse begins on one direction. Once one direction has curled the rest follow, so the grouped directions collapse together. It has two halves: the walls (each fold makes the next easier) and the counting (with interchangeable points the fully curled, symmetric state weighs more). Both halves are in the toy's arithmetic (Sec. VI). T34 tests the first and T42 the second.

Walls, named points (T34), measured. Flat $6 \times 6 \times 6$ ($N = 216$) and $8 \times 8 \times 8$ ($N = 512$) tori at $\lambda = 1.25$. Energies run from 64, the cheapest move out of flat space, to one and two directions' worth of curling cost plus a wall (260 and 500 at $N = 216$; 560 and 1,060 at $N = 512$). The energy is packed into one vertex's store or spread through $2N$ stores; eight replicas per cell; 50,000 sweeps. All 20 cells have a MELTED majority. No replica ended with a single folded vertex, none reached a whole direction's worth of folded vertices at any reading, and

none melted first and folded later. Verdict: MELTS. The author's prediction, RE-CURLS, one direction first and then the rest, failed. Ours, MELTS, held.

What those MELTED states are. In an exploratory quench, with its expectation written first, all 80 final graphs at $N = 216$ were run for 5,000 sweeps with only moves that do not raise the energy. All 80 came to rest on exactly flat space, with no defect left. By the record's later reading, made with T42's below, the MELTED replicas end with one to a few small scars, 12 to 46 vertices of 216 or 512, most of them one swapped pair of links, flickering on and off. The largest excitation, about 150 of 512 vertices at the largest energy, healed back to such scars. Nothing curled. The verdict stands by its rule and says less than its word: the energy excites flat space, and the excitation heals [1].

Counting, interchangeable points (T42), measured. Flat $8 \times 8 \times 8$ at $\lambda = 1.02$, where curling all three directions of the whole torus costs 123 units. The spread protocol with 64 to 2,048 units; eight replicas per cell; 50,000 sweeps; and a control with the same chain and the automorphism factor off (named points). Interchangeable points: HEALS, with 47 of 48 replicas HEALED and one MELTED at the largest energy. Named control: MELTS, with a MELTED majority in five of six cells. The author's prediction, inferred for this design by the assistant, was RE-CURLS with interchangeable points and MELTS in the control. The first half failed and the second held. Ours, MELTS in both with no visible difference, failed for the interchangeable half.

The named control did not melt (a correction on the record). Its MELTED replicas hold 12 to 42 vertices of 512 above the flat signature at the last reading: one to a few scars, most of them 12 vertices holding 64 units, one swapped pair of links, which is the cheapest move out of flat space. At the smaller energies the scar switches on and off about 40 times a run and is present about three quarters of the time, so the last reading is a snapshot of a flicker. At the largest energy up to about half the vertices are disordered early on, with either kind of point, and heal back. At most 2 of 512 vertices ever fell below the flat signature. Both treatments heal. With named points a small scar keeps flickering; with interchangeable points it almost never forms. *Interpretation (ours, unverified):* a scar at each of its many possible places is a different named arrangement but one interchangeable arrangement, so the scar has far fewer ways to exist. Here the counting works against the scar. It does not work for the fold.

V. FOUR DIRECTIONS

The author's claim (26 September 2026). Time curls behind the present: the present is a moving front, and a state with one direction curled lies below flat four-direction space. The toy has no time. It can ask only whether flat eight-link space re-curls from a local push when a one-curled state lies lower.

Measured (T47, part B). Flat $6 \times 6 \times 8 \times 8$ ($N = 2{,}304$) at $\lambda = 1.30$, with a tie table: an extra cost per vertex that depends on its number of open directions. Under the test table the one-curled state lies 0.24 per vertex below flat space; under the control table flat space is lowest. The cheapest move out of flat space costs 128 under both. One push of 128.01 units was placed in one vertex's store; three replicas per table; 20,000 sweeps. The registered rule: MELTED if a quarter or more of the vertices are above $d = 4$, FRONT if a quarter or more are at $d = 3$, HEALS if fewer than 2% are anywhere but $d = 4$, STALLED otherwise. Verdict: HEALS under both tables. At most 0.9% of the vertices ever left $d = 4$, and the runs end with a damaged patch of about 20 vertices that holds the push and neither spreads nor curls. The author's prediction, inferred by the assistant and confirmed by her before any result was read, was FRONT under the test table and HEALS under the control. The first failed and the second held. Ours, HEALS under both, held. *Interpretation (ours, unverified):* a patch of 20 vertices gains about 5 units from lying lower, far less than its boundary costs, so a re-curled region would have to be large before it grew by itself. A larger or more concentrated push was not tested. The table's values came from a fit to an order of openings that the author has since set aside [1]; that does not change what the run tested.

VI. WHAT IS EXACT

The floor. Written link by link, Eq. (1) at four links is $H = 4\sum_e[(2 - S_e) + \lambda(S_e - 2)_+]$, and a link carries at most three squares. At $\lambda \geq 1$ every term is nonnegative, so nothing lies below the flat sheet. Flat space is stable there, not metastable. On the six- and eight-link ladders flat space is the lowest rung at $\lambda > 1$.

No first step. Enumerating every switch the chain can make out of a perfect 10×10 or 12×12 sheet: none adds a square; 14,900 and 34,128 lose squares; the cheapest costs 32. Out of a disordered graph 5,127 switches add a square, and the cheapest of them is downhill by 75. So a fold cannot begin in one step from a flat sheet. One order can still be built from another through a thin seam where squares are briefly broken, as the opening of the curled torus shows in reverse [2]. In the runs above the broken squares did not go on to fold.

Price does not decide it. At $\lambda = 1.25$ a square added to a flat sheet costs 4 and a square destroyed costs 16, so per square a fold is the cheaper damage by a factor of four. In the exploratory runs of Sec. III the folded share of the damage rose as λ fell toward 1, to 23% at most at $\lambda = 1.05$ against 10% at most at 1.25, and never reached one half.

The wall of flat space. The cheapest move out of flat space costs 32, 64 and 128 at four, six and eight links. It loses 2, 4 and 8 squares and adds no surplus square, so

it does not depend on λ.

The walls' half of the mechanism. On the six-link ladder, on the tori and the gas of 6-cubes computed, the cheapest single move out of a state with one, two and three curled directions costs $96-48\lambda$, $96-64\lambda$ and $96-80\lambda$: 36, 16 and downhill at $\lambda = 1.25$. The barrier between two rungs is the same from either side, so each fold lowers the excess cost of the next by 16λ. Nothing in the untied energy ties the directions together. *Interpretation (ours, unverified):* a single move need not be the whole barrier in three directions, where the boundary of a curled patch can grow with the patch. That barrier is priced in a separate registered calculation whose verdict is not yet read [1, 4].

The counting's half. With interchangeable points the free energy of an arrangement is $H - g \ln A$. At $N = 512$, $A = 12{,}288$ for flat $8 \times 8 \times 8$, 2,048 with one direction curled, 12,288 with two, and 3.2×10^{39} for the fully curled state, a gas of eight 6-cubes. The partial folds carry no more symmetry than flat space; only the fully curled state gains. At $g = 1.5$ it beats flat space only for $\lambda < 1.020$ (1.013 at $g = 1.0$). Identical cubes can be exchanged, so the crossing rises slowly with their number: 1.031 at 10^3 cubes, 1.045 at 10^6, and 1.25 only near 10^{52}. This compares single arrangements and ignores their thermal excitations. T42 ran at $\lambda = 1.02$, and nothing folded.

VII. THE AUTHOR'S TIE, AND THE TEST UNDER IT

The author's decision (9 October 2026). The three directions are tied, "all at the last." A vertex with d of its three directions open pays an extra $f(d)$, with $f = (0, a, 2a, 0)$ for $d = 0, 1, 2, 3$. She regards the untied runs as a stepping stone. The toy with this tie is our family, not CQG.

Exact under the tie. Every rung of the six-link ladder sits $3a$ per vertex above flat space, so nothing is released until a vertex's last direction opens. Read as a fold: curling one direction of flat space costs $3a$ per vertex, the same as curling all three. The first fold pays the whole price and the later folds are free. The wall of flat space stays 64 at every λ computed, 1.05 to 1.25. As implemented, a vertex above the flat signature pays no tie, so the tie charges curling and not breaking.

The author's mechanism for the fold (her words, 9 October). A high amount of energy concentrates in space, then more arrives, until "there is just too much and it has to go somewhere and the only mechanism it can go to is curling macrodimensional space itself."

Under the tie (T57), measured. None of the six tests above used this tie. T57 is T34's protocol on flat six-link tori of 216 and 512 vertices, with the tie and with an untied control, at $\lambda = 1.10$ and 1.25, packed and spread, eight replicas per cell and 50,000 sweeps, at energies up to 3,200 units at $N = 512$: about twice what curling one direction of the whole torus costs under the tie at $\lambda = 1.25$. The reading rule is T34's, with a verdict for each pair of tie and λ. Verdict: MELTS in all four groups. Of 640 replicas 582 are MELTED, 58 HEALED and none FOLDED. No replica reached a whole direction's worth of folded vertices; the most vertices below the flat signature at any reading is 15 with the tie and 25 without. The author's prediction, as the registration reads her words, was RE-CURLS with the tie at the largest energies. It failed. Ours, MELTS at $\lambda = 1.25$ and MELTS or HEALS at 1.10, tied and untied alike, with no FOLDED replica, held.

Where the energy went. Into broken squares. Up to about one direction's worth, a replica ends with a median of 12 damaged vertices, one swapped pair of links, as in T34. Packed into one store at one direction's worth and at twice it, a median of 97 to 107 and then 255 to 263 of 512 vertices are damaged at the end, the same with the tie as without. Spread through a bath the same energies leave 12 to 79. At the largest packed energies MELTED means what it says by the count: half the torus above the flat signature. Those end states' wiring has not been read. *Interpretation (ours, unverified):* the energy had another place to go. As implemented the tie charges a curled vertex and not a broken one, so it makes breaking the cheaper of the two; and with or without it no single move out of flat space adds a square. Two exact counts would separate these and are not done: the price of the smallest curled patch in flat space against the same number of broken vertices, and the same under a table that charges damage. Such a table would be a new rule, and it is the author's to choose [1].

What a fold would need (interpretation, ours, unverified). Before anything in the toy is called a black hole it would have to show, in order: a stable, localized curled region inside open space; a reproducible threshold; growth with stored energy; trapping or an analog of it; and a sensible large-scale limit. With four links one such region is known, the relic left when a curled torus opens: one curled column, a local minimum costing $24\lambda - 16$ that lasts only in the cold [3]. With six links none has been found: not among one- and two-switch constructions (exact within that search, at $\lambda = 1.02$), and not as a remnant of heat damage in the quench of Sec. IV. The toy also has no pull: at fixed wiring, defects that share no square do not interact through the energy [10]. Nothing in it gathers energy, so the energy is placed by hand. Where the entropy attributed to a black hole would live in an ordered interior is unanswered.

VIII. WHAT THIS DOES NOT SHOW

It shows nothing about real black holes. It is a statement about a graph toy with no time and no pull, at 36 to 512 vertices (2,304 with eight links, on three replicas per table), over sweeps that are not a physical clock.

No registered test here is at $\lambda = 1$, so no verdict here is a result in CQG.

The packed protocol was not run with interchangeable points in three directions, and the tie runs with named points only.

MELTED is a count. *Interpretation (ours, unverified):* where the wiring was read, in three directions at these energies, the toy produced neither the author's fold nor a lasting disordered bubble; flat space was excited and healed. In two directions the largest spread budget of the exploratory run left every vertex melted by the count, as heating at equilibrium does by construction. Whether that state is the random phase that Ref. [5] reads as a black hole's interior was not measured. A correction of the author's stands beside this: a region returned to the random phase is not obviously the opposite of her picture, and may be a different account of the interior.

Every six- and eight-link statement carries the open reproduction gate of Sec. II. No physicist has reviewed this work.

ACKNOWLEDGMENTS

Code, analysis and text were drafted with Claude (Anthropic), with the 9 October scope review assisted by Codex (OpenAI), under the author's direction; the author has not yet read this revision, and no physicist has reviewed it. The repository holds every configuration, seed, result and pre-registration: `https://github.com/EmilySmithCreate/RecreationalPhysics`.

[1] `https://github.com/EmilySmithCreate/RecreationalPhysics`: `docs/papers/series_plan.md`; PREREGISTRATION T21, T26, T27, T34, T42, T47 (part B), T57; ASSUMPTIONS O20, O21, O22, O25, O36, O41, O43, O48, O49, O50, O51, O53, O55, O58, O64, O66, O69, O72, O75, O80, O82, O96, O103, O106, O114, Q12, Q18, Q20, Q21, Q22; VISION Updates 5, 19, 23, 24, 25, 27, 36, 47, 50, 51, 54.

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